

F-2. Multilayer Plantation				
Forest / Shrub Land Conservation	High	Medium		
Measure Type	Cost	Preferences by Community	Hazards	Implementation Time of Measures

Descriptions									
Measures	Multi-layer farming is a type of polyculture that involves growing multiple crops in a single area. It generally involves growing plants at different heights, so that one crop grows on the ground and another grows above it. This technique is often employed in plantations and orchards to maximize solar energy use, even when there are many plants present. In hill tracts, multilayer plantations are a type of agricultural practice where diverse tree, shrub, and crop species are planted at varying vertical levels to maximize land utilization in the difficult terrain. Taller trees, such as fruit trees or timber species, are planted in the top layer of this system, followed by medium-sized trees, medicinal plants, or spice crops in the middle layer, and low-growing crops, such vegetables or ground coverings, in the bottom layer. This approach not only enhances the productivity of the land but also helps in conserving soil and water. In the Chattogram Hill region, where soil erosion and steep slopes are significant issues, multilayer plantations successfully reduce erosion by stabilizing the soil with an array of root systems. The growth of underneath crops is encouraged by the shade that higher trees' canopies provide to the lower strata. By increasing food security and diversifying their revenue streams, this approach helps local farmers while maintaining environmental sustainability in the area's vulnerable ecosystem.								
	Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77
Applicable Sites	• Terraced & Contour Farming Areas • Erosion-Prone Degraded Lands								
Slope	Less than 3%		3% to 15%		15% to 30%		30% to 60%		Above 60%
			√						
Design options and performance	• Tree-Based Multi-Layer System can be introduced which Includes timber or fruit trees (top layer), medicinal/spice plants (middle layer), and vegetables/ground cover crops (bottom layer). • Agroforestry Multi-Layer Farming, which Integrates forestry species with food and cash crops for sustainable land use. • Contour Strip Cropping – Aligns plants along contour lines to reduce runoff and increase water infiltration. • Silvi-Horticulture System – Combines fast-growing tree species with fruit and vegetable crops for economic benefits.								

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	• Forest-Garden Model – Encourages biodiversity through diversified tree and shrub species.			
Feasibility criteria	Technical Feasibility • Ensures efficient vertical layering and light utilization. • Uses contour planting and mulching to reduce erosion. • Maintains soil fertility and water retention. Environmental Feasibility • Prevents erosion and land degradation through sustainable water flow management. • Maintains ecological balance by minimizing habitat disruption. • Restores native flora and fauna habitats. Economic Feasibility • Uses locally available seedlings and community labor. • Generates income through agroforestry, medicinal plants, and non-timber forest products. • Improves land productivity, reduces climate risks, and enhances local economies Social Feasibility • Engages local farmers in planning, implementation, and maintenance. • Enhances food security and economic opportunities. • Aligns with government policies and local governance structures for long-term sustainability.			
Operation and maintenance	• Soil Conservation & Fertility Management – Uses organic manure, mulching, and crop rotation. • Tree Pruning & Canopy Management –Optimal light penetration for lower layers should be ensured. • Weed & Pest Control – Promotes biological pest control and intercropping. • Seasonal Crop Adjustments – Rotates shade-loving crops in the lower layers based on seasonal variations. • Community-Based Monitoring – Involves farmers in training and land management activities.			
Cost	High			
Benefits				
	Type	Low	Medium	High
	Canopy Cover Increase			
	Carbon Sequestration			
	Crop Production Increase			
	Crop Failure Risk Reduction			
	Ensuring Food Security			
	Fruit Production Increase			
	Forestry Goods Production Increase			
	Livestock Production Increase			
	Fisheries Production Increase			
	Alternative Livelihood Generation			

	Increase of Income			
	Soil Erosion Control			
	Improved Soil Health			
	Soil Fertility Increase			
	Sustainable Water Supply			
	Water Quality Improvement			
	Water Purification			
	River Bank Erosion Control			
	Surface Runoff Reduction			
	Groundwater Recharge			
	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			
	Better WASH facilities			
Implementation Challenges				
	Type	Low	Medium	High
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	• Soil Erosion Reduction – Strengthens land stability through multi-root systems. • Improved Carbon Sequestration – Increases carbon absorption through tree-based systems. • Sustainable Water Use – Enhances rainwater absorption and reduces surface runoff. • Biodiversity Conservation – Supports natural ecosystems and wildlife corridors.			
Remarks	Effective agroforestry in the CHT depends on site-specific planning, ensuring that species selection and arrangement are carefully tailored to local soil and climate conditions. Thorough community training is crucial, equipping farmers with comprehensive knowledge of crop choices, planting density, and ongoing maintenance requirements.			